Meeting Minutes

Subject:	Meeting with Krishna Ragil from Haemograph
Group name/ identifier	Group 24
Meeting Location, Date & Time	Online on Microsoft Teams Date – 10 September 2026, Time – 12:00 pm to 1:00 pm
Group members present	Jianhao Zhang, Ria Mishra, Rohan Roy Mathew, Xingcan Wang, Xuan Zhang
Apologies	

1 Actions from previous meetings

Task	Who is responsible	Deadline
Continue exploring and analysing the available datasets	Team	Before meeting
Implement and test the proposed change-point detection approach	Team	Before meeting
Compare detected change points with established rheological criteria	Team	Ongoing

2 Items discussed and decisions made

Item	Decision (if applicable)
Change-point parameter selection	The basis for selecting parameter ranges and thresholds should be clearly justified rather than chosen by trial and error. The selection should be supported by literature, data characteristics or a systematic sensitivity analysis.
Significant change-point identification	Where multiple change points are detected, the analysis should determine which point is most meaningful. Feature or peak prominence and other quantitative criteria were suggested as possible approaches.
Open-source clotting dataset	An open-source human plasma clotting dataset containing seven healthy donors and 63 trauma patients was presented. Differences between groups were observed, but their cause cannot be assumed because experimental and timing conditions may also contribute.
Reference clotting time	A practical reference time was defined by fitting a flat baseline before a candidate split and a rising line after it, then selecting the split with the minimum total squared error. This was treated as an operational reference rather than a true biological onset.
Method comparison	STFT, WTMM and Gaussian-based approaches were compared using sensitivity, specificity, balanced accuracy and mean absolute timing error. Different methods showed different strengths, so classification performance and timing accuracy should be considered separately.

Noise robustness and validation	Performance decreased as noise increased. Further work should clarify exactly how each method generates its predicted clotting time and validate the framework across individual samples before making generalised conclusions.

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3 Agreed actions from this meeting

Task	Who is responsible	Deadline
Provide a clear justification for parameter and threshold selection, supported by sensitivity analysis and/or literature	Team	Before next meeting
Clarify in writing how each detection method identifies or predicts the clotting/change-point time	Team	Before next meeting
Send the presentation and follow-up explanation of the method outputs to the supervisor	Team	As soon as possible

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